

ETIENNE PHILLIPS

<https://etiennephillips.com>

NSF GRFP Fellow, Barry M. Goldwater Scholar, Ronald E. McNair Scholar

E-mail: ephil@berkeley.edu; Citizenship: US Citizen

EDUCATION

University of California, Berkeley <i>PhD, Mathematics, Expected 2029</i> Advisors: Thomas Scanlon, Kevin Carlson	<i>2024 - present</i>
North Carolina State University <i>B.S. Mathematics (Honors), Minor in Philosophy</i>	<i>2021 - 2024</i> GPA: 4.00

PUBLICATIONS

- “Hausdorff measure of zeros of nontrivial multivariable polynomials,” *Bulletin Des Sciences Mathématiques*. (2025)
with K. Nguyen, A. Murdza
- “On ranked and bounded Kohnert posets,” *Order* (2025)
with L. Colmenarejo, F. Hutchins, and N. Mayers; [arXiv:2309.07747](https://arxiv.org/abs/2309.07747)
- “Multivariate Fibonacci-like Polynomials and Their Applications,”
with S. Park, P. Qi, I. Ziba, and Z. Zhan; [arXiv:2309.08123](https://arxiv.org/abs/2309.08123)

PAID RESEARCH EXPERIENCES

University of Maryland, College Park REU in Mathematics Principal Investigator: Christian Zickert	<i>2023</i>
A quantitative analysis of Sard’s Theorem Principal Investigator: Khai Nguyen, funded by NSF	<i>2022-2023</i>
Poset Structure of Kohnert Diagrams Principal Investigator: Laura Colmenarejo, funded by NC State and NSF	<i>2022-2023</i>
Michigan State University SURIEM REU Principal Investigator: Aklilu Zeleke	<i>2022</i>

AWARDS AND HONORS

Graduate

NSF GRFP Fellow	<i>2024 - 2029</i>
PKP Graduate Bond Fellowship	<i>2024</i>

Undergraduate

College of Sciences Outstanding Transfer Student	<i>2024</i>
Churchill Scholarship Nominee	<i>2023</i>
College of Sciences Student Excellence Award	<i>2023</i>
Barry M. Goldwater Scholar	<i>2023</i>
Ronald E. McNair Scholar	<i>2022 - 2024</i>

TALKS AND PRESENTATIONS

Research Talks

<i>On Kohnert Posets that are Lattices</i>	2024
Joint Mathematics Meeting, AMS Contributed Paper Session on Algebraic Structures and Related Topics	
<i>Quantitative Analysis of Sard's Theorem</i>	2023
Math Honors Talks, NC State University	
<i>Multivariate Fibonacci-like Polynomials and Their Applications</i>	
Young Mathematicians' Conference, The Ohio State University	2022

Seminar, Expository, and Other Talks

<i>Lawvere Duality for Lawvere Algebraic Theories</i>	2025
UC Berkeley Categorical Semantics Seminar	
<i>Introduction to Derived Categories</i>	2025
UC Berkeley Geometric Representation Theory Student Seminar	
<i>The Beauty of the Yoneda Lemma</i>	2023
NC State, Mathematical Insights	
<i>Why There's No Such Thing as a Crazy Belief</i>	2022
TEDxNCState	

TEACHING EXPERIENCE

GSI for Math 1B / Math 51: Calculus	<i>Spring 2025, Summer 2026, Fall 2026</i>
UC Berkeley	
TA for MA425: Mathematical Analysis	<i>Spring 2022, 2023</i>
NC State University	
Professional Tutoring	<i>2020 - 2022</i>
Various locations including TC Tech Prep, Smartboost Learning, VarsityTutors.com, and NC State TRIO programs.	

LEADERSHIP AND MENTORSHIP POSITIONS

Mentor, UC Berkeley Directed Reading Program in Mathematics	
Arjun Garg, Simplicial and Infinity Categories	2026
Theo Danial, Monads in Category Theory	2026
Austin Lin, Double Categories	2026
Xichen Song, Applied Category Theory	2026
Ailsa Sun, Algebraic Topology and Category Theory	2025
Mentor, Educational Opportunity Program (EOP) STEM	2026
Chris Chen, UC Berkeley	
Mentor, Compass Program Grad Student	<i>2025-2026</i>
Alex Gao, Austin Lin, and Muer Zhou, UC Berkeley	
President, Society For Undergraduate Mathematics	<i>2022 - 2024</i>
NC State University	
Organizer, Putnam Competition Club	<i>2021 - 2024</i>
NC State University	

VOLUNTEER WORK

Housing Options for Students Today (HOST)

2021 - 2022

Volunteer

HOST is a nonprofit program designed to alleviate student homelessness in Wake County. I help promote the organization by interviewing with local news outlets like Raleigh Magazine, CBS 17, Spectrum News, etc.

Habitat for Humanity Student-Led Group

2017 - 2019

President

Raised \$13,750 to build a home. Coordinated a team of dozens of volunteers to organize fundraisers such as bake sales, Nickels for Nails, etc., as well as volunteer builders for construction.